



Lab Task:	14
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Course Code:	CL2005
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Course Title:	Database Systems - Lab
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Semester:	Spring 2026
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Guidelines

- Please review the lab manual before starting the task to ensure you fully understand the requirements.
- You may use **only the concepts covered in the lab manual** to complete this task.
- Your submission must be a **single PDF file** that includes:
 - The source code (in plain text)
 - Screenshots of the corresponding outputs
- Name your file using the following format: **rollno_name_labtasknumber.pdf**
Example: *24p-1234_ozair_labtask14.pdf*
- Unethical use of AI tools will result in a **deduction of 5 marks**.
- Late submissions will incur a penalty of **-1 mark per day**.

Tasks

You are provided with a JavaScript file named *lab-14_15-devs.js*, which contains MongoDB commands to create and populate two collections: *developers* and *projects*.

Execute/Import the given script file in MongoDB shell to load the dataset. After successfully setting up the database, perform the following tasks:

1. Display the name, country and experience of developers who are **not from Pakistan** and whose experience is between 3 and 10 years.
2. Find developers who either:
 - have "JavaScript" in their skills
 - OR have experience greater than 7
 - AND whose current project exists
3. From the projects collection, display the title, domain, and budget of projects where:
 - technologies include either "Node.js" or "Docker"
 - AND budget is greater than 5000



4. From the projects collection, find projects where:
 - team_size is between 2 and 5
 - AND technologies array contains exactly 2 elements
5. Display the name and skills of only the first 2 developers who **do not have MongoDB in their skills** AND whose experience is less than or equal to 5.
6. Display developers where:
 - current_project is **missing or null**
 - OR number of skills is 2
7. Find the **top 3 most experienced developers** who have experience ≥ 3 years.
8. Display the name, skills and experience of developers whose skills include **either "Python" or "Go"**, but NOT "Java", and whose experience is greater than 4.
9. Display the name, project name and budget of developers whose **current project budget is greater than 10000 AND status is "active"**.
10. Display the name, skills, and project budget of developers who have **both "JavaScript" and "MongoDB"** in their skills and whose current project budget is between 4000 and 10000.